



Research Paper

Efficient jet-assisted single-phase immersion liquid cooling for high heat-flux servers

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ABSTRACT

With the trend toward integration and decarbonization in data centers, exploring clean and efficient cooling solutions has become increasingly urgent. In response, this study proposes an innovative single-phase immersion liquid cooling (SPILC) system with jet-assisted enhancement. By establishing a three-dimensional cooling model, the performance of conventional and jet-assisted SPILC systems is compared. Moreover, an in-depth study is conducted on the impacts of different jet designs and heatsink optimization on jet-assisted SPILC systems. The results show that the thermal management performance of jet-assisted SPILC systems has significantly improved compared to the conventional one, with the electronic component temperature and coolant temperature uniformity index of jet-assisted SPILC systems decreasing by 6.1 % and 73.1 %, respectively. For the jet-assisted SPILC system, the coolant temperature uniformity is better in the horizontal layout, but the flow resistance is increased compared to the vertical layout. Moreover, its cooling performance improves with the increased jet flow rate ratio, which achieves the best at a jet angle of 90 °. Balancing the trade-off between flow resistance and heat transfer, the thermal management performance of jet-assisted SPILC systems is maximized when the jet-impacted high heat-flux device's heatsink has a pin fin size of 3 × 3 mm.

1. Introduction

With the boom of big data, artificial intelligence, and 5G, data centers (DCs) have emerged as essential technical support for developing the digital economy [1]. However, equipment in server rooms under high load generates a large amount of heat, which seriously affects the stable and safe operation of DCs. To maintain reliable thermal management, the power consumption of DC cooling equipment has already reached over 30 % of the total electricity consumption [2]. Given this, developing clean and efficient cooling systems has become a necessary trend in the construction of next-generation DCs.

Different from conventional air-cooling approaches, existing efficient heat transfer methods include single-phase liquid cooling [3–5], heat pipes [6–8], and phase change thermal storage [9–11]. In particular, single-phase liquid cooling technology has attracted widespread attention due to its strong heat transfer capability, simple structure, and high controllability advantages. Moreover, this technology can be classified as indirect and direct two ways depending on whether the work material is in direct contact with heat-generating devices [12]. Among them, the liquid cooling plate (LCP) is one of the leading indirect cooling

methods in practical engineering applications. It achieves relatively efficient heat dissipation by closely attaching the LCP to the electronic device [13]. Currently, related works are mainly focused on the structural and competence optimization of microchannel LCPs. Miao et al. [14] proposed two new LCPs with variable cross-section channels and found that both new LCPs are superior to conventional LCPs with rectangular channels in terms of cooling ability. Qu et al. [15] established mathematical models for DCs using the LCP cooling and obtained functions of power consumption and chip temperature, achieving energy savings of 42.7 % under variable heat loads and reducing the power usage effectiveness (PUE) to 1.16. However, due to the need for special leak detection instruments and rack-level pipeline systems, LCP cooling requires customized design according to the actual equipment layout of DCs, reducing system universality and increasing investment costs [16]. Moreover, the possibility of coolant leakage caused by hot-plugging during the maintenance process increases, and issues of safety and reliability are also highlighted [17]. In contrast, immersion liquid cooling employs dielectric fluids as coolants, all of which exhibit excellent insulating properties and chemical stability [16]. Furthermore, in the immersion liquid cooling system, the coolant directly contacts heat-generating components to carry away heat, not only enhancing

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Nomenclature			
A	area of the inlet or outlet	β	flow rate ratio of the mainstream to the jet stream
c_p	specific heat capacity	ε	turbulent kinetic energy dissipation rate
C_μ	turbulent viscosity correction parameter	λ	thermal conductivity
$C_{1\varepsilon}, C_{2\varepsilon}$	model constant	μ	dynamic viscosity
F_p	pressure force loss	ν	kinematic viscosity
g	gravitational acceleration	μ_t	turbulent viscosity
G_k	turbulent kinetic energy produced by the average velocity gradient	ρ	density
$\bar{h}$	averaged convective heat transfer coefficient	$\sigma_k, \sigma_\varepsilon$	model constant
k	turbulent kinetic energy	σ_T	temperature uniformity index
L_{CPU}	characteristic length	Subscripts	
n	number of coordinate points	in	inlet
N	mesh number	in1	mainstream inlet
$\overline{Nu}$	local Nusselt number	in2	jet stream inlet
$\bar{p}$	average pressure	max	maximum
Q_{in}	inlet flow rate	out	outlet
Re	Reynolds number	Abbreviations	
$\bar{T}$	average temperature	CPU	central processing unit
T'	temperature pulsation	DC	data center
T_{avg}	mean temperature of coolants	DIMM	dual inline memory module
T_{in}	inlet temperature	LCP	liquid cooling plate
T_0	initial temperature	PCB	printed circuit board
$\bar{u}_i, \bar{u}_j$	average velocity component	PCH	platform controller hub
u_i', u_j'	velocity pulsation	PSU	power supply unit
v	velocity	PUE	power usage effectiveness
x_i, x_j, x, y, z	coordinate axis	SPILC	single-phase immersion liquid cooling
Greek symbols		TPILC	two-phase immersion liquid cooling
α	coefficient of thermal expansion	3D	three-dimensional

heat transfer capacity and improving the long-term reliability of the equipment but also making the cooling system simpler and more compact. Based on whether the phase transition of coolants occurs during the circulation process, immersion cooling is classified into single-phase immersion liquid cooling (SPILC) [18–21] and two-phase immersion liquid cooling (TPILC) [7,22–24]. For the SPILC system, Wang et al. [25] compared its thermal-flow-energy performance of high heat-flux servers with other cooling approaches through numerical simulation. Compared to forced air cooling and LCP schemes, the SPILC system significantly reduced the highest temperature and temperature gradient of high heat-flux components and greatly improved the thermal transport uniformity, reducing PUE by 17.6 % to 20.8 %. Wang et al. [26] proposed a new TPILC system and studied the cooling performance of lithium batteries, comparing natural, forced air cooling, mineral oil (SPILC), and SF33 fluid (TPILC) four cooling schemes. Their findings illustrated that the TPILC system has the best cooling performance under all test conditions. Although TPILC is superior to SPILC in terms of cooling performance, it is comparatively inferior in cost, security, and maintainability. Thus, SPILC systems are considered even more promising for large-scale commercial deployment.

Current research on SPILC primarily emphasizes analyzing the flow behaviors and thermal transport characteristics, the performance assessment and coolant selections, and the system control and optimizations. Huang et al. [27] compared the overall performance of buoyancy-driven and pump-driven SPILC systems. The numerical simulation results indicate that the average temperature and PUE of the pump-driven SPILC system decreased by 55.5 % and 11.6 %, respectively. Then, their team researched the performance of SPILC systems through experiments, focusing on the impact of two opposing coolant flows [28]. Those results indicated that anti-gravity flows decrease the peak temperature, heat resistance, and PUE by 33.8 %, 55.6 %, and 1.4

%, respectively, compared to those SPILC systems with pro-gravity flows. Importantly, their investigation reveals that using the coolant with low viscosity can improve the cooling performance and energy consumption. Inspired by this, many studies have delved into the relationship between coolant thermal properties and SPILC performance. Chen et al. [29] developed a Figure of Merit as an evaluation metric for the coolant in SPILC systems. Moreover, they studied the impact mechanism of different thermal properties on the SPILC performance through experiments and simulations. The results show that the thermal properties are ranked as viscosity > heat conductivity > density > specific heat capacity according to the importance of Figure of Merit. In addition, optimizing the control and structure of SPILC systems is also a hot topic. Through numerical simulation, Liu et al. [30] researched the effects of different cabinet structures and perforated plates on the SPILC performance. The results imply that using a perforated plate could improve the flow uniformity but might increase pressure loss. Ni et al. [31] modeled a SPILC system using FC-40 and studied the impact of hole radius, inlet arrangement, and hole count. Those findings illustrate that a narrower radius favors balancing the flow in manifold holes by increasing flow resistance. Shrigondekar et al. [32] researched the thermal transfer characteristics of SPILC systems under different working conditions. Their experiments present that when the coolant entry temperature is raised from 15 °C to 35 °C, the heat resistance decreases while the server temperature increases.

A comprehensive literature review reveals that current enhancements in SPILC performance primarily rely on passive methods, which do not result in substantial performance gains. To address this limitation, pioneering studies have proposed a solution that enhances SPILC system performance by integrating it with active jet-assisted heat transfer enhancement techniques. For jet-assisted SPILC systems, Qi et al. [33] constructed a three-dimensional (3D) numerical model to

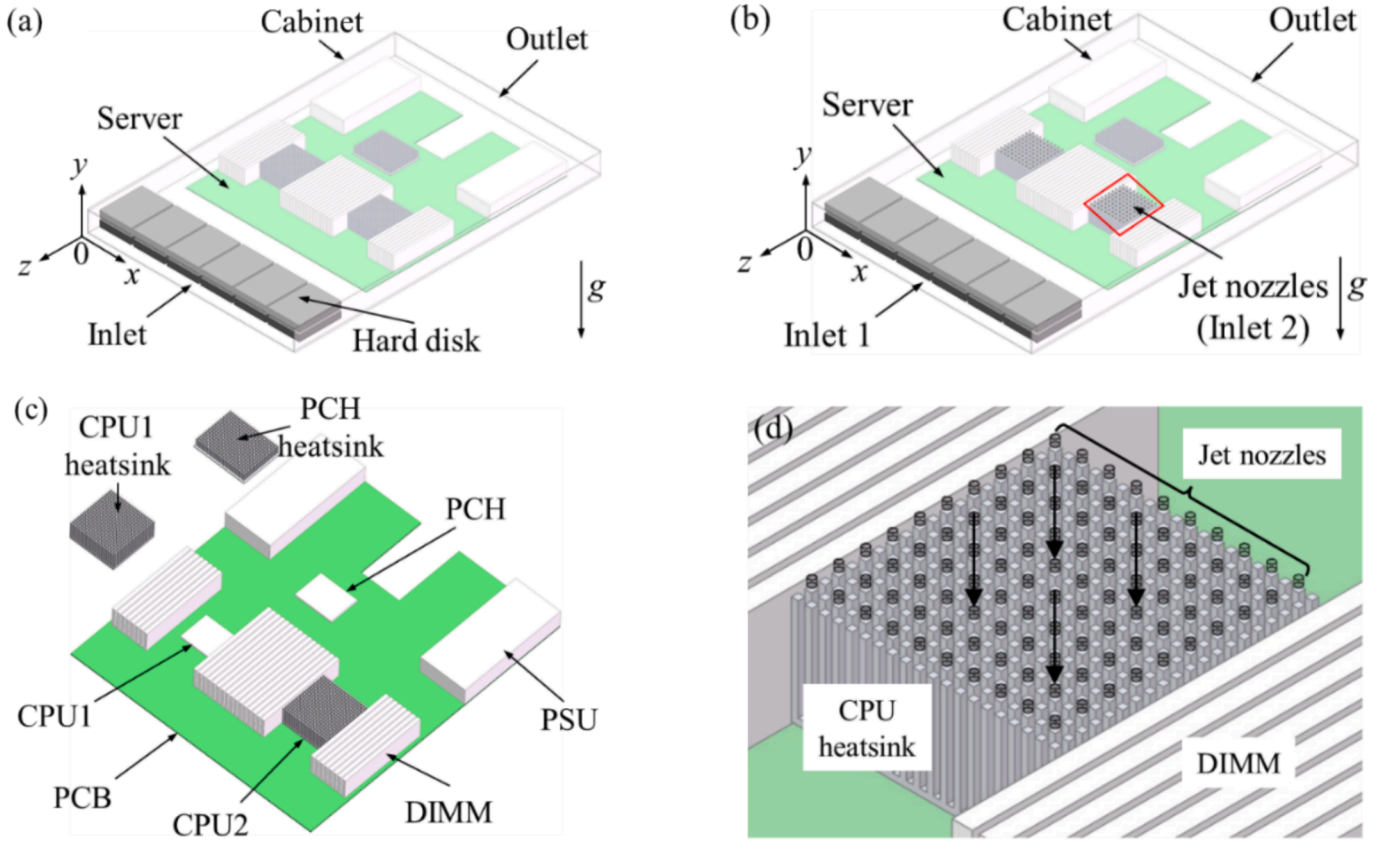


Fig. 1. Schematic diagram of the two SPILC cabinets. (a) Conventional SPILC cabinet; (b) Jet-assisted SPILC cabinet; (c) Server motherboard; (d) Jet nozzles.

study their heat transfer performance for the thermal management of avionics electronic server modules under various hole diameters and distances. The results indicate a significant enhancement in the performance of the jet-assisted SPILC system. The best local heat transfer effect is achieved when the hole diameter is 5 mm and the hole distance is 10 mm. Additionally, Xue et al. [34] conducted an in-depth study on the jet-assisted SPILC for a single-chip heat sink, focusing on the analysis of nozzle diameter, spacing, and distribution. The results reveal that the overall cooling performance is optimal with a nozzle diameter of 2 mm, and adopting a non-uniform nozzle layout could greatly improve temperature uniformity compared to a uniform nozzle arrangement. These findings highlight the exciting potential of jet-assisted SPILC systems to revolutionize heat transfer performance. However, there is a significant research gap in the existing literature related to jet-assisted SPILC systems. This gap is particularly important given the potential advantages of jet-assisted SPILC systems in terms of flow resistance, heat transfer characteristics, and energy efficiency. Understanding the impact mechanism of flow and heat transfer performance in jet-assisted SPILC systems, as well as the underlying enhancement mechanisms, is therefore a pressing need. In this context, urgent and extensive research is needed in this field to fully understand and harness the potential of jet-assisted SPILC systems.

To address the abovementioned critical issues, this study develops a novel jet-assisted SPILC system for high heat-flux DC servers. By establishing 3D steady-state numerical models for conventional and jet-assisted SPILC systems, an in-depth analysis is conducted on the flow and heat transfer characteristics within different SPILC systems. Moreover, this study also examines the impact of layout, flow rate ratios, and jet angles on the jet-assisted SPILC system. In addition, the structural optimization design of the CPU heat sink on the jet-assisted SPILC server is carried out to provide practical data support and technical guidance for the practical application of SPILC systems in high heat-load DCs.

Table 1

Dimensions and thermal power of components on the server motherboard.

Component name	Dimensions (mm)	Power (W)	Quantity
CPU	50 × 50 × 3	375	2
DIMM	144 × 35 × 6	24.2	32
PCH	65 × 50 × 3	39	1
PSU	183 × 80 × 30	175.68	2
Hard disk	100 × 75 × 15	11.25	12

2. Numerical methodology

2.1. Geometry description

To quantitatively investigate the flow and thermal transport characteristics of jet-assisted SPILC systems, a simplified server geometric

Table 2

Thermophysical properties of various materials.

Properties	Symbols	FC-3283 (25°C) [35]	Silicon	Copper	FR 4
Density (kg/m ³)	ρ	1820	1900	8978	1200
Dynamic viscosity (mPa·s)	μ	1.4	/	/	/
Specific heat capacity [J/(kg·K)]	c_p	1100	795	381	880
Thermal conductivity [W/(m·K)]	λ	0.066	120	387.6	orthotropic
Coefficient of thermal expansion (1/K)	α	0.0014	/	/	/

based on Intel's M50CYP1UR server is constructed. Fig. 1(a) and (b) present the conventional SPILC cabinet and the jet-assisted SPILC cabinet, respectively. Both cabinets have a computational domain with dimensions of 730 mm (length) $\times$ 488 mm (width) $\times$ 45 mm (height), and operate in a front-in and rear-out coolant flowing mode. The coolant used herein is the electronic fluorinated liquid from 3M (Product number: FC-3283). Fig. 1(c) presents a schematic diagram of the server motherboard. Components within the cabinet include the CPU, power supply unit (PSU), dual inline memory module (DIMM), platform controller hub (PCH), heatsink, printed circuit board (PCB), and hard disk. The dimensions and thermal power of each component are listed in Table 1. The material for components is silicon, the heatsink is made of copper, and the PCB is made of FR 4. Table 2 gives the thermal properties of these materials. The jet-assisted SPILC cabinet is equipped with an array of jet nozzles above the CPU heatsink, as shown in Fig. 1(d). These nozzles, with a default diameter of 2 mm, cover the area directly above the CPU heatsink, with 100 nozzles per CPU (10 rows $\times$ 10 columns), resulting in an open area ratio of 5.16 %. The CPU heatsink is a rectangular pin-fin heatsink with the default pin-fin cross-sectional size of 2 $\times$ 2 mm, a height of 30 mm, and a base thickness of 2 mm.

2.2. Governing equations

To investigate the performance of different SPILC systems, this study has established a 3D steady-state flow and heat transfer mathematical model. Notably, the following assumptions have been made to simplify the calculations.

- (1) The coolant flow within the computational domain is steady and incompressible.
- (2) In this study, the conventional SPILC cabinet has an inlet Reynolds number (i.e., $Re = 5872.4$) greater than 4000, and the jet-assisted SPILC cabinet has a jet inlet Reynolds number (i.e., $Re = 3487.1$) greater than 2300 in the default state. Therefore, the flow regime is turbulent.
- (3) Thermal radiation is neglected.
- (4) Wall heat dissipation and viscous losses are neglected.
- (5) The Boussinesq hypothesis is employed for constructing a link between the Reynolds stress and the mean velocity gradient.

Considering the above assumptions, the flow and heat transfer mathematical model in SPILC cabinets can be described by the following equations.

Continuity equation:

$$\frac{\partial(\bar{u}_i)}{\partial x_i} = 0 \quad (1)$$

Momentum equation:

$$\rho \frac{\partial(\bar{u}_i \bar{u}_j)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_j} (\bar{u}_i \bar{u}_j') \quad (2)$$

Energy equation:

$$\rho c_p \frac{\partial(\bar{u}_i \bar{T})}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} - \rho c_p \bar{u}_i \bar{T}' \right) \quad (3)$$

where $\bar{u}_i$, $\bar{u}_j$ denote the average velocity component, $\bar{T}$ is the average temperature, $\bar{p}$ represents the average pressure, ν is the kinematic viscosity, u_i' , u_j' are the velocity pulsation in different directions, and T' denotes the temperature pulsation.

To solve the conservation equations mentioned above, this study intends to use the Standard k - ϵ model, as referenced in the previous literature [36,37], to describe the flow characteristics within the SPILC system. It offers good robustness, cost-effectiveness, and reasonable predictive capabilities for a wide range of turbulent flows. Therefore, the standard k - ϵ model has been widely popular in engineering for fluid flow and heat transfer simulations.

k equation:

$$\rho \frac{\partial(k \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right) + G_k - \rho \epsilon \quad (4)$$

ϵ equation:

$$\rho \frac{\partial(\epsilon \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (5)$$

where k represents turbulent kinetic energy, ϵ denotes turbulent kinetic energy dissipation rate, G_k denotes the turbulent kinetic energy produced by the average velocity gradient. Moreover, μ_t denotes the turbulent viscosity, which can be expressed by

$$\mu_t = \rho C_\mu \frac{k^2}{\epsilon} \quad (6)$$

where C_μ denotes the turbulent viscosity correction parameter (i.e., $C_\mu = 0.09$). Other model parameters are: $\sigma_k = 1.0$; $\sigma_\epsilon = 1.3$; $C_{1\epsilon} = 1.44$; $C_{2\epsilon} = 1.92$.

In this study, the initial temperature within the computational domain is:

$$T|_{t=0} = T_0 \quad (7)$$

Inlet boundary conditions are:

$$T_{in} = T_0 \quad (8)$$

where T_0 represents the initial temperature, T_{in} denotes the inlet temperature of coolants, both have a value of 30 °C. For the conventional SPILC cabinet, the inlet flow rate of coolants (i.e., Q_{in}) is a constant. Moreover, the total inlet flow rate of a jet-assisted SPILC cabinet is given by

$$Q_{in} = Q_{in1} + Q_{in2}; \beta = Q_{in1} / Q_{in2} \quad (9)$$

where the subscripts *in1* and *in2* denote the inlet 1 and inlet 2 of the jet-assisted SPILC cabinet, respectively (as shown in Fig. 1(b)), and β

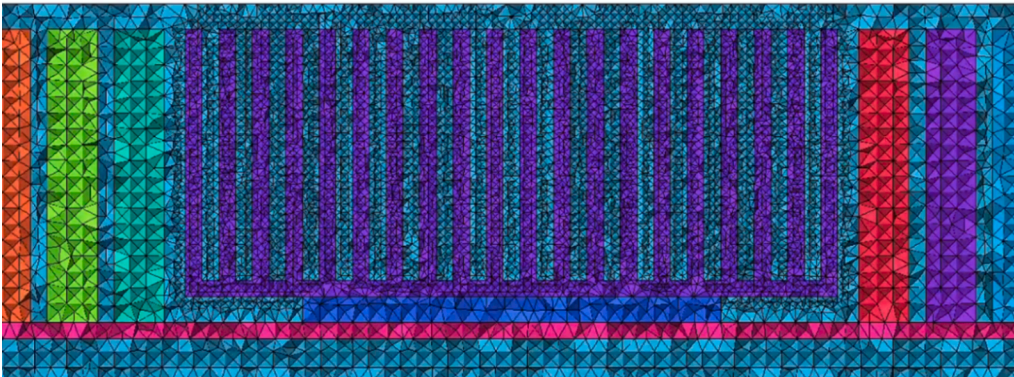


Fig. 2. Schematic diagram of the mesh section of the CPU.

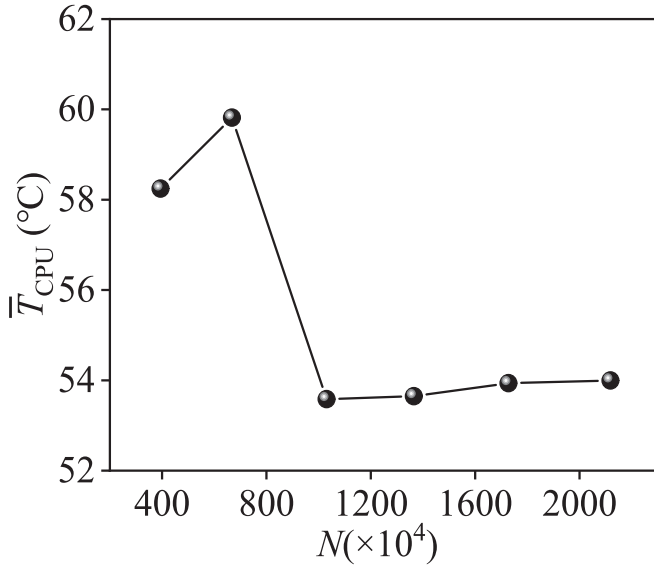


Fig. 3. Mesh independence verification.

denotes the flow rate ratio of the mainstream to the jet stream. By default, the total flow rate of the jet-assisted SPILC cabinet is the same as the inlet flow rate of the conventional SPILC cabinet, both set at 70 L/min. Notably, this study aims to investigate the impact of jet angle on the performance of jet-assisted SPILC cabinets. Moreover, the jet angle in this study is the angle between the jet flow direction and the main flow direction. Additionally, the outlet boundary is defined as outflow, the outer wall surface is adiabatic, and the liquid–solid interface is characterized by no-slip boundaries along with conjugate heat transfer conditions.

2.3. Solution method and independence tests

To investigate the performance differences between conventional and jet-assisted SPILC systems, this study conducts numerical computations using the mainstream computational fluid dynamics software ANSYS Fluent. The flow behaviors and thermal transport characteristics within the SPILC cabinet were simulated using the finite volume method. The ICFM CFD package from ANSYS software was employed to generate mesh for different SPILC cabinets. Given the complexity of the geometric structure, this study opts for a widely used type of unstructured grid, namely Tetra/Mixed. Specifically, this study generates a volume mesh primarily consisting of hexahedral elements within the computational domain and creates a prism boundary layer mesh. It is worth mentioning that the boundary layer mesh comprises three layers, incorporating CPU, heat sink, and other liquid–solid coupling wall surfaces. Fig. 2 illustrates a schematic diagram of the mesh section of

typical devices. For the pressure–velocity coupling, a SIMPLE-based pressure correction method was chosen. The pressure term used the widely applicable PRESTO scheme, and the momentum and energy equations were discretized using a second-order upwind scheme. The corresponding relaxation factors were adjusted to 0.2 for pressure, 0.8 for momentum, and 0.75 for energy. The numerical solutions were thoroughly checked for accuracy and reliability, and a double-precision solver was selected. Finally, the numerical computation was considered converged, and the iterative calculation was stopped when the residuals of the energy equation and other equations were less than 10^{-6} and 10^{-4} .

Ensuring the reliability and reproducibility of the numerical simulation results is a key goal of this study. To achieve this goal, this study validates mesh independence using the jet-assisted SPILC cabinet as an example. Six different mesh systems are selected, and the operating conditions for all cases are set to default. Specifically, in the jet-assisted SPILC cabinet with a horizontal layout, the flow rate ratio between the mainstream inlet and the jet stream inlet is 3/7, and the jet angle is 90° . Fig. 3 shows the variation curve of the average CPU temperature inside the jet-assisted SPILC cabinet under different computational grids. It can be observed that these values remain almost constant after the mesh number exceeds 13,651,051. This finding guides us to use a minimum of 13,651,051 meshes in all numerical calculations, striking a balance between computational cost and accuracy.

2.4. Model validation

To verify the simulation effectiveness, this study selects the SPILC experiment by Huang et al. [28] for comparative analysis. Fig. 4(a) shows the physical model of the SPILC cabinet, including components such as CPU, PCH, DIMM, etc. Notably, the materials, coolant properties, and geometric dimensions used in the model constructed herein are consistent with those in the literature [28]. Moreover, the simulation calculations are performed using the same boundary conditions as in the literature [28], with the coolant inlet temperature at 15°C and the flow direction being anti-gravity. Fig. 4(b) gives the maximum monitored temperature of the CPU at different volumetric flow rates (2 L/min ~ 8 L/min) during the simulation and experimental processes. Obviously, the temperature data from the numerical simulation matches the experimental data of Huang et al. [28], with an error range between 0.43 % and 2.86 %. The sources of error are mainly the experimental measurement errors and the deviation caused by simplifying heat sources to uniform heating bodies in these simulations, and the error range is reasonable for scientific research.

3. Results and discussion

To quantitatively elucidate the factors affecting the SPILC characteristics, this study introduces the dimensionless Nusselt number to represent the heat convection intensity. Notably, here refers explicitly to the local Nusselt number near the CPU, which is given by

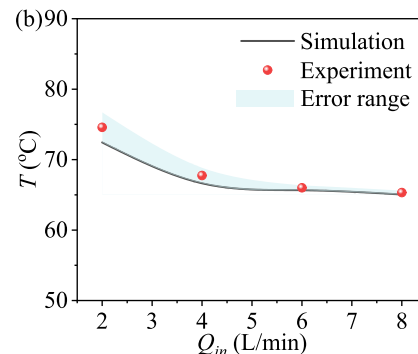
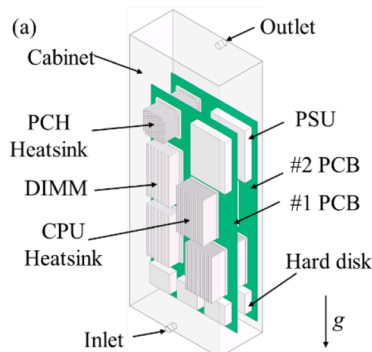


Fig. 4. Model validation for the conventional SPILC system. (a) Physical model; (b) Comparison of simulations and experiments.

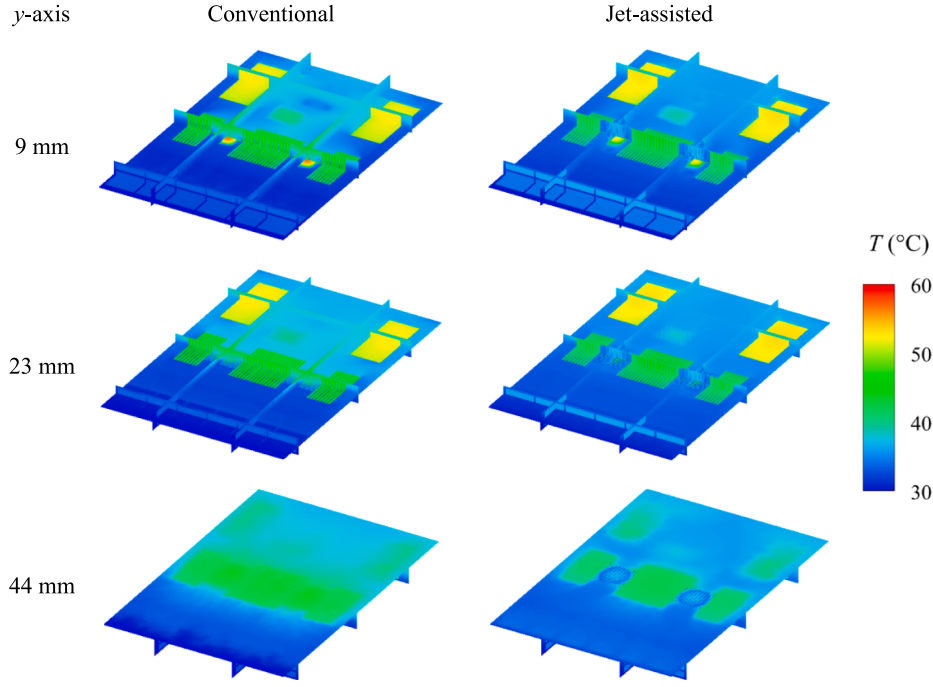


Fig. 5. Temperature distribution in two SPILC cabinets.

$$\overline{Nu} = \frac{\overline{h}L_{CPU}}{\lambda} \quad (10)$$

where $\overline{h}$ is the averaged convective heat transfer coefficient, L_{CPU} is the characteristic length corresponding to the CPU and its heatsink.

To quantitatively evaluate the thermal uniformity within SPILC cabinets, this study also introduces a temperature uniformity index (i.e., σ_T), defined as the standard deviation of the temperature values at various coordinate points in the coolant. Analogous to the function of standard deviation representing the dispersion of a dataset, the temperature uniformity index in this study is used to characterize the temperature distribution uniformity in the coolant region.

$$\sigma_T = \sqrt{\frac{1}{n} \sum_{i=1}^n (T_i - T_{avg})^2} \quad (11)$$

where n denotes the number of coordinate points, and T_{ave} denotes the mean temperature of coolants. According to the definition, a smaller σ_T indicates better temperature uniformity of coolants [25].

Considering that the jet-assisted SPILC system has two inlets, it is therefore not appropriate to characterize its flow resistance loss using the common pressure drop. Thus, this study introduces an evaluation index of pressure force loss (i.e., F_p) to assess the energy loss of the coolant flowing through different SPILC cabinets.

For the conventional SPILC cabinet:

$$F_p = (p_{in} - p_{out})A \quad (12)$$

For the jet-assisted SPILC cabinet:

$$F_p = p_{in1}A_{in1} + p_{in2}A_{in2} - p_{out}A_{out} \quad (13)$$

where the subscripts *in* and *out* denote the inlet and outlet of the SPILC system, respectively. A denotes the area of the inlet or outlet of a SPILC cabinet.

3.1. Comparison of conventional and jet-assisted SPILC systems

To explore the differences in the heat transfer mechanisms of two different SPILC cabinets, this study selects temperature fields for

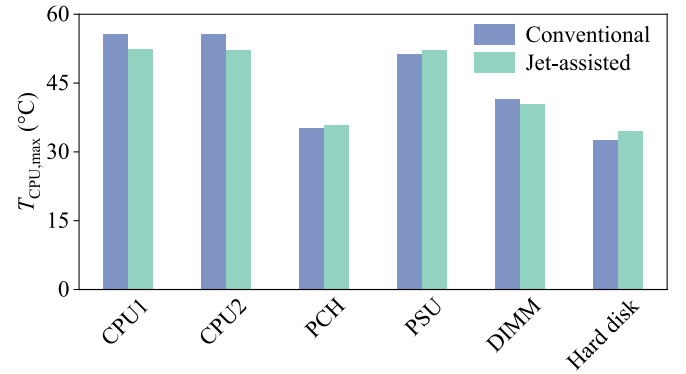


Fig. 6. Peak temperatures of key components in two SPILC cabinets.

analysis at two x-axial sections ($x = 139.5$ mm, $x = 348.5$ mm), three z-axial sections ($z = 70$ mm, $z = 346$ mm, $z = 590$ mm), and three y-axial sections ($y = 9$ mm, $y = 23$ mm, $y = 44$ mm) of the heat-generating components. In this section, the boundary and initial conditions are default values for all cases. Furthermore, the jet-assisted SPILC cabinet is arranged horizontally with a flow rate ratio of 3/7 and a jet angle of 90° . As observed from Fig. 5, the temperature gradient at the CPU section ($y = 9$ mm) is higher than that at other server components ($y = 23$ mm) and the coolant section ($y = 44$ mm). Compared with the conventional SPILC cabinet, the components within the jet-assisted SPILC cabinet are at lower temperatures, particularly in the vicinity and downstream of CPUs, indicating that the jet-assisted SPILC performs better. Moreover, the temperature distribution of coolants in the jet-assisted SPILC cabinet is more uniform. This is because part of the coolant impacting the area near the CPU heatsink as a jet, enhancing heat dissipation in the area with high heat flux density around the CPU. Additionally, the average velocity of the coolant in the jet-assisted SPILC cabinet is faster under the same volume flow rate, leading to more excellent flow disturbance and further enhancing heat transfer.

Fig. 6 presents the peak temperatures of the main components within different SPILC cabinets to quantitatively analyze the cooling

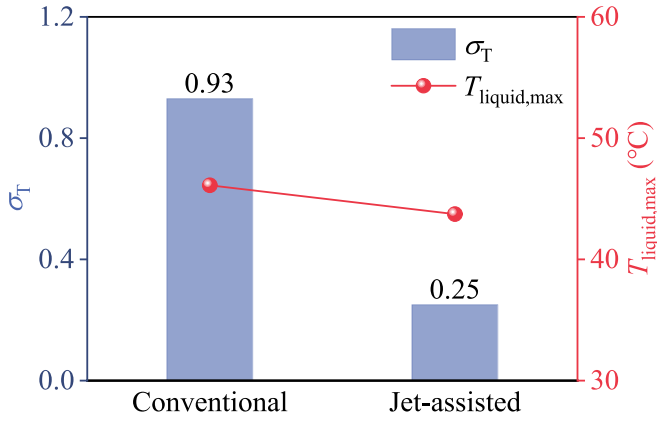


Fig. 7. Temperature uniformity index and peak temperature of coolants in two SPILC cabinets.

performance enhancement. As observed from the figure, the peak temperatures of the CPU, PCH, and DIMM in the jet-assisted SPILC cabinet are significantly lower than the conventional one. Quantitative analysis reveals that the peak temperatures of CPU1 and CPU2 are decreased by 6.0 % and 6.1 %, respectively. A close observation indicates that the components with reduced temperatures in the jet-assisted SPILC cabinet are located near the jet nozzles and in their downstream areas. However, the hard drives upstream of the jet nozzles and the PSU located on the side experience a slight deterioration in heat transfer due to the reduced coolant flow. Given this, with the same total volume flow rate, distributing more coolant to the high heat-flux CPU through the jet nozzles

decreases the CPU temperature and improves the cooling ability of components affected by the thermal barrier in the vicinity and downstream of the CPU.

Fig. 7 plots the coolant temperature uniformity index and peak temperature within different SPILC cabinets. It should be noted that the temperature uniformity index is calculated based on the temperature values at 32 different coordinate points within the SPILC cabinet ($x = 20, 140, 350, 470$ mm; $y = 15, 35$ mm; $z = 150, 260, 450, 600$ mm). These points are arrayed throughout the coolant to ensure a uniform representation of the coolant's temperature distribution. As observed from Fig. 7, the coolant temperature uniformity index in a jet-assisted SPILC cabinet has decreased by 73.1 % compared to the conventional one, indicating a significant improvement in temperature uniformity. The peak temperature has decreased by 5.2 %. This is attributed to the jet's effective role in reducing the temperature in the high heat-flux area of the cabinet, particularly near the CPU, thereby reducing the overall thermal cascade effect and making the temperature distribution within the cabinet more uniform.

To research the flow behaviors of different SPILC cabinets, this study analyzes the velocity fields at the same cross-sections mentioned above. It is evident from Fig. 8 that the coolant cross-section ($y = 44$ mm) in both SPILC cabinets has a smaller velocity gradient compared to the CPU cross-section ($y = 9$ mm) and other server component cross-sections ($y = 23$ mm) due to the absence of obstruction from the server motherboard. Moreover, the velocity distribution does not exhibit significant abrupt changes. The cloud map at the CPU cross-section ($y = 9$ mm) shows that the coolant velocity in the bypasses on both sides of the conventional SPILC cabinet is significantly faster than that in the jet-assisted SPILC cabinet under the same total coolant volume flow rate.

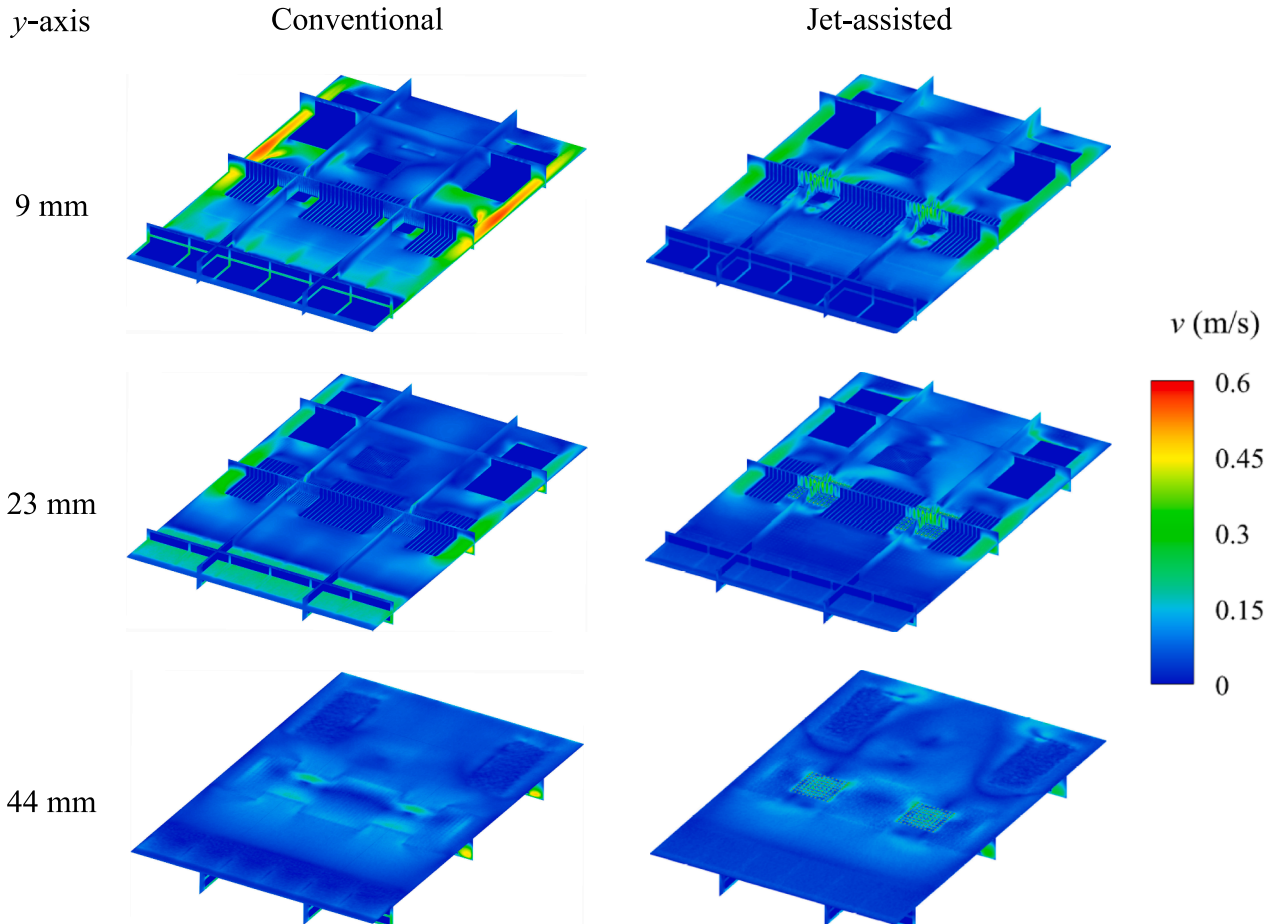


Fig. 8. Velocity distribution in two SPILC cabinets.

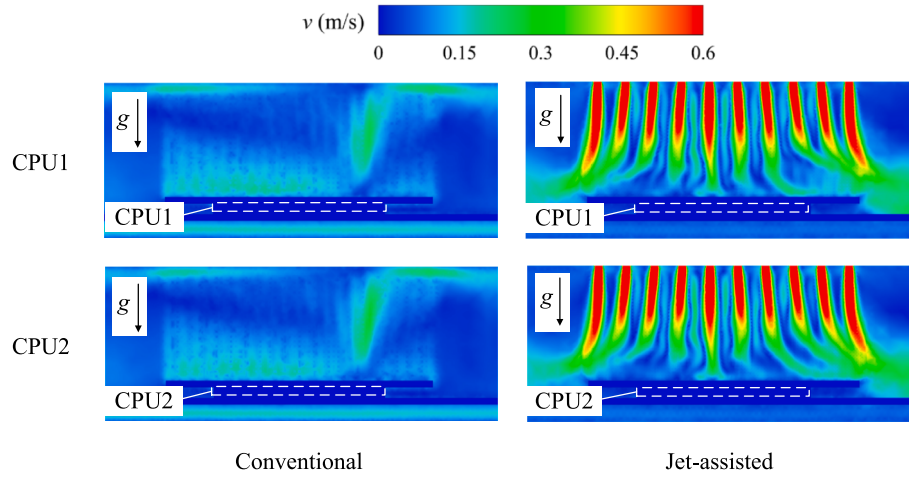


Fig. 9. Local flow field evolution near CPU in two SPILC cabinets.

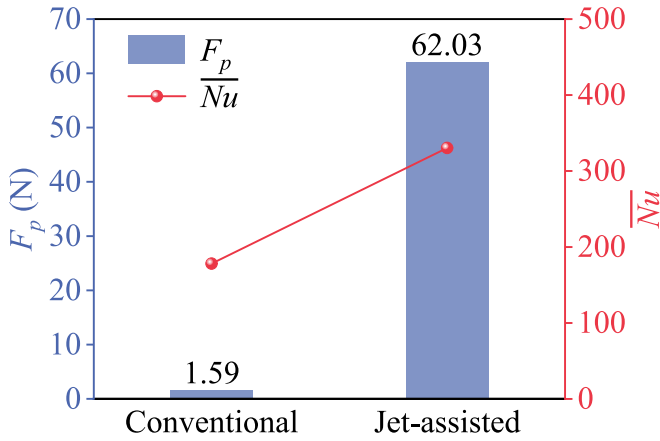


Fig. 10. Pressure force loss and local Nusselt number in two SPILC cabinets.

This phenomenon implies that a large portion of the coolant in the conventional SPILC cabinet flows into the bypasses, resulting in a smaller effective volume flow rate and slower velocity of coolants directed towards the CPU, thereby leading to heat transfer deterioration in the CPU and its downstream areas.

To compare the local flow characteristics around the CPU and heatsink in different SPILC cabinets, this study selects two sets of local cross-sections along the x-direction ($x = 135.5, 352.5$ mm) of the CPU and heatsink for analysis. Fig. 9 presents that the velocity distribution in different local areas of the same SPILC cabinet is approximately the same. For the jet-assisted SPILC cabinet, the coolant jet passes through the gaps of the heatsink and impacts the base, enhancing the heat

dissipation effect. In contrast, the velocity in the CPU heatsink area of the conventional SPILC cabinet is smaller, while the velocity in the bypass section between the PCB board and the lower boundary is larger, indicating that a considerable amount of coolant flows into the bypass behind the PCB board for the conventional SPILC cabinet. Compared to the conventional one, the velocity within the CPU heatsink of the jet-assisted SPILC cabinet is significantly higher, and the amount of coolant flowing to the back of the PCB board is less, resulting in a markedly enhanced local cooling effect for the CPU.

Fig. 10 presents a comparison of the cabinet pressure force loss and the local Nusselt number of CPU1 for two different SPILC cabinets. As observed from the figure, the pressure force loss of the jet-assisted SPILC cabinet is significantly increased compared to the conventional SPILC cabinet, indicating that the flow resistance within the jet-assisted SPILC cabinet is significantly larger than that of the conventional one. This is because a considerable portion of the coolant in the jet-assisted SPILC cabinet enters through the inlet 2, while the volume flow rate entering through the inlet 1 is relatively reduced, resulting in a significant reduction in flow resistance upstream of the inlet 2 and an overall reduction in flow resistance. Moreover, the local Nusselt number of CPU1 in the jet-assisted SPILC cabinet is higher than that in the conventional SPILC cabinet, increasing by 85.5 %. This further demonstrates that the heat convection performance within jet-assisted SPILC cabinets is better than that of the conventional one. This is attributed to the greater flow rate and faster velocity of the coolant near the CPUs and its heatsinks in the jet-assisted SPILC cabinet, which effectively enhances the intensity of convective heat transfer.

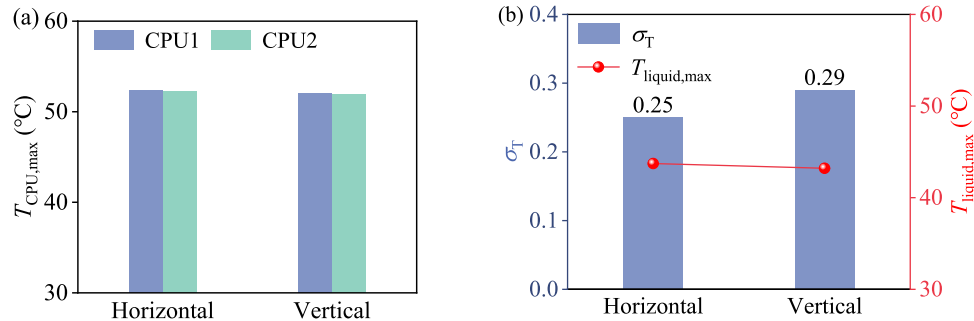


Fig. 11. The impact of SPILC cabinet layout on the temperature characteristics. (a) Peak temperature of CPUs; (b) Temperature uniformity index and peak temperature of coolants.

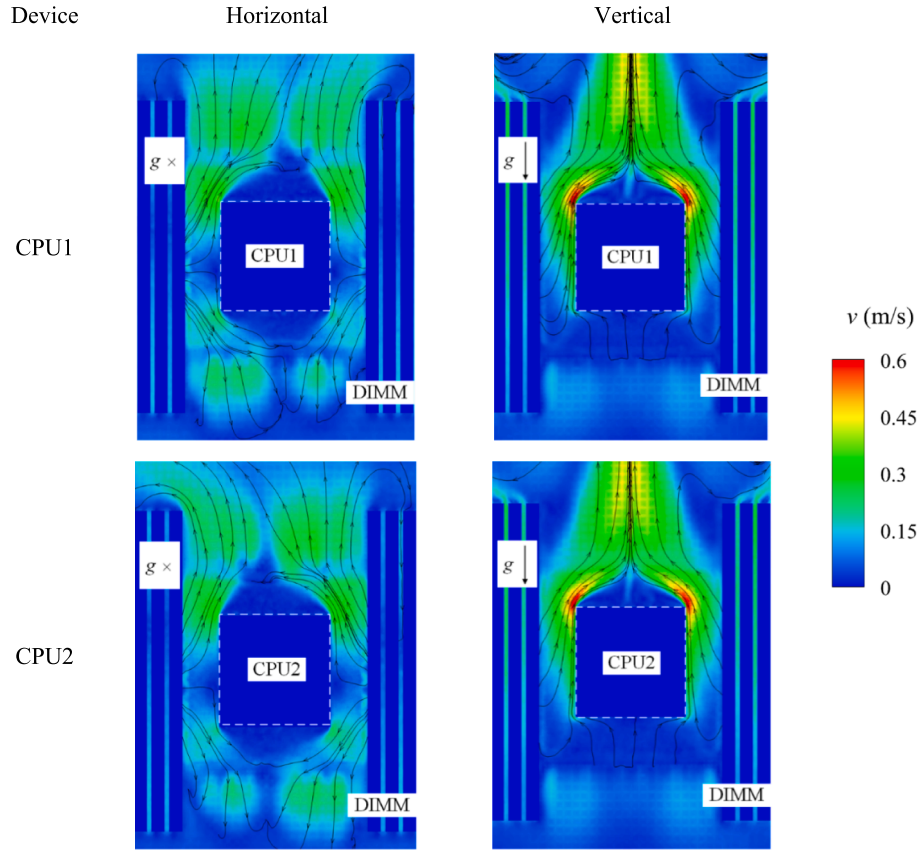


Fig. 12. Velocity distribution of CPU local area under different cabinet layouts.

3.2. Performance influence of jet-assisted SPILC cabinets

To explore the performance-influencing factors of jet-assisted SPILC cabinets, this study has conducted research on the impacts of cabinet layout, flow rate ratio, and jet angle on the cooling performance.

3.2.1. Influence of cabinet layout

Fig. 11 illustrates the internal temperature response characteristics under horizontal and vertical SPILC cabinet layouts. In this section, the flow rate ratio is 3/7, and the jet direction is perpendicular to the CPU. Fig. 11(a) shows that compared to the horizontal layout, the peak temperatures of CPU1 and CPU2 in the vertical layout only decreased by 0.31 °C and 0.4 °C, respectively. This indicates that the impact of cabinet layout on CPU temperature response is minimal. Moreover, Fig. 11(b) illustrates that the coolant temperature uniformity index in the horizontal SPILC cabinet is 13.8 % lower than that in the vertical one, while the maximum coolant temperature increased by 0.5 °C. Given this, the

CPU cooling performance in the horizontal SPILC cabinet is not significantly improved, but its coolant temperature uniformity is better than in the vertical SPILC cabinet. This is because the layout direction mainly affects the natural convection within the SPILC cabinet. For the horizontally arranged SPILC cabinet, the main flow direction is perpendicular to gravity, and the temperature distribution in the flow direction is less affected by natural convection. However, in the vertically arranged SPILC cabinet, the main flow direction is against gravity, and there is a significant height difference between the inlet and outlet. In this context, natural convection is fully developed, leading to obvious temperature stratification of the coolant in the upper and lower parts, reducing temperature uniformity. Furthermore, more coolant is drawn into the bypass between the PCB board and the lower boundary due to gravity in the horizontal layout, reducing the coolant flow towards the CPU. This results in a slight reduction in thermal transport and a minor rise in the peak temperatures of both the CPU and coolant.

Fig. 12 presents a comparison of the flow characteristics around the

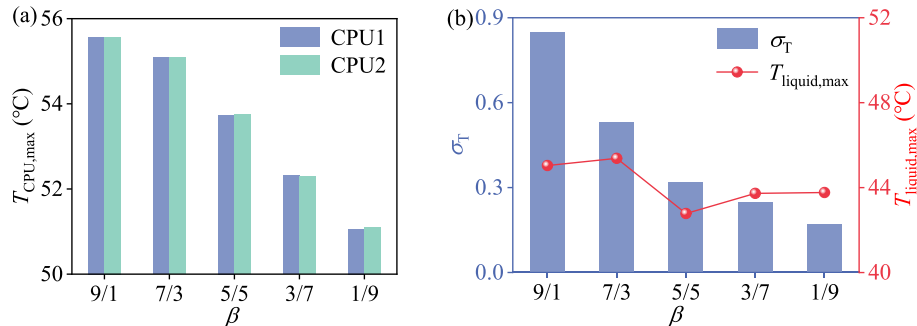


Fig. 13. The impact of flow rate ratio on temperature characteristics. (a) Peak temperature of CPUs; (b) Temperature uniformity index and peak temperature of coolants.

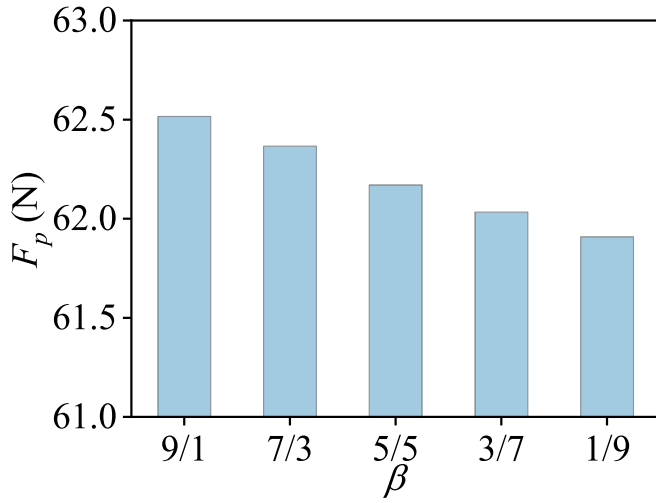


Fig. 14. The impact of flow rate ratio on the pressure force loss.

CPU in SPILC cabinets with different layouts. The velocity fields at the CPU's local cross-sections along the y-direction ($y = 9$ mm) are depicted. The figure reveals that vortices occur upstream and downstream of the CPU in the horizontally arranged cabinet. A distinct backflow phenomenon is also observed downstream in the main flow direction, leading to increased flow resistance and reduced coolant velocity near the CPU, which in turn weakens convective heat transfer. The reason for this is the jet's impact on the CPU heat sink, causing it to spill around and disrupt the main flow, leading to the formation of vortices and local backflow. In contrast, the jet's direction in the vertically arranged cabinet is perpendicular to the gravitational force, resulting in less impact on the main flow.

3.2.2. Influence of flow rate ratio

By delving into the impact of the flow rate ratio in jet-assisted SPILC cabinets on flow behaviors and thermal transport characteristics, this study provides valuable insights for practical applications. Fig. 13 quantitatively compares the effects of different flow rate ratios (i.e., $\beta = 9/1, 7/3, 5/5, 3/7$, and $1/9$) on the temperature characteristics inside the jet-assisted SPILC cabinet. In this section, the jet-assisted SPILC cabinet is horizontally arranged with a jet angle of 90° . As shown in Fig. 13(a), the peak CPU temperature gradually reduces as the jet flow rate increases, a finding that can be directly applied to enhance the cooling efficiency of electronic devices. Fig. 13(b) indicates that the coolant temperature uniformity index within the SPILC cabinet decreases with an increasing jet flow rate ratio, a result that can guide the

design of more uniform temperature distribution systems. The peak temperature of the coolant also shows a decreasing trend, and it gradually stabilizes when the jet flow rate exceeds half of the total, providing practical guidance for system stabilization. This is also due to the increased jet flow, which strengthens the cooling effect around the CPU, reducing their thermal cascade effect and leading to a more uniform temperature distribution overall.

Fig. 14 quantitatively plots the impact of the flow rate ratio on the pressure force loss of jet-assisted SPILC cabinets. It can be observed from the figure that the pressure force loss is negatively correlated with the flow rate ratio. As the proportion of jet flow increases, the flow resistance within jet-assisted SPILC cabinets gradually reduces. This phenomenon demonstrates that a rise in the jet flow rate not only contributes to the cooling performance improvement, but also reduces the flow resistance loss of jet-assisted SPILC cabinets.

3.2.2.1. Influence of jet angle. Fig. 15 analyzes the impact of jet angle on the temperature response characteristics inside the SPILC cabinet. In this section, the jet-assisted SPILC cabinet is horizontally arranged with a flow rate ratio of $3/7$. Fig. 15(a) shows that the peak CPU temperature first reduces and then rises as the jet angle increases from 30° to 150° , reaching its minimum at a jet angle of 90° . This practical insight can guide the design and optimization of cooling systems, ensuring the best cooling effect. Fig. 15(b) indicates that both the temperature uniformity

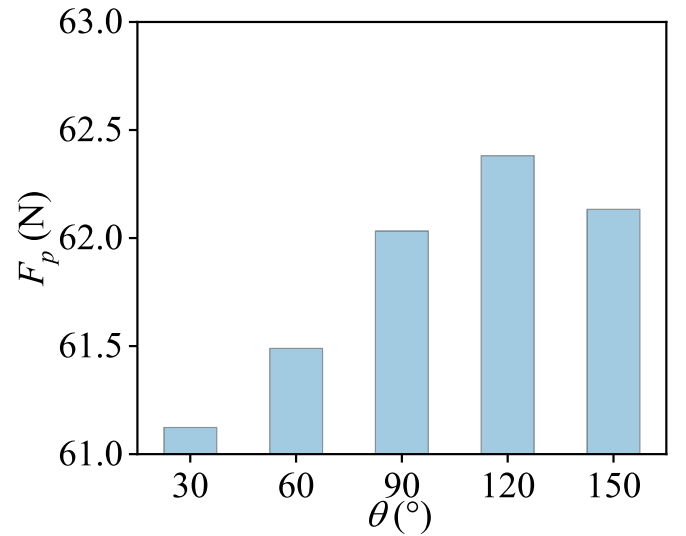


Fig. 16. The impact of jet angle on the pressure force loss.

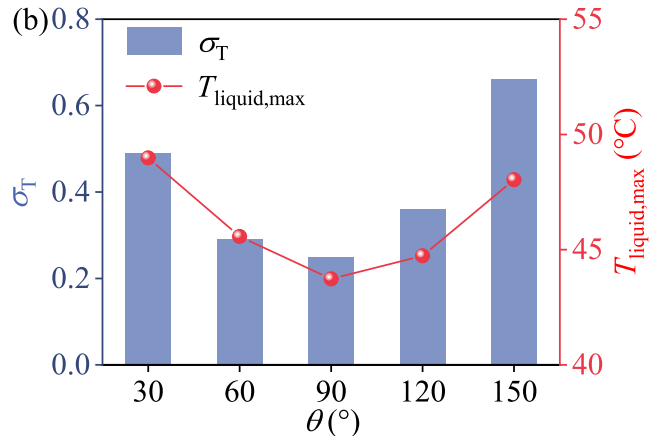
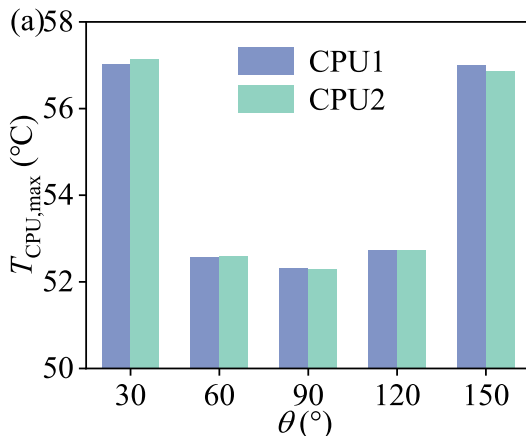


Fig. 15. The impact of jet angle on temperature characteristics. (a) Peak temperature of CPUs; (b) Temperature uniformity index and peak temperature of coolants.

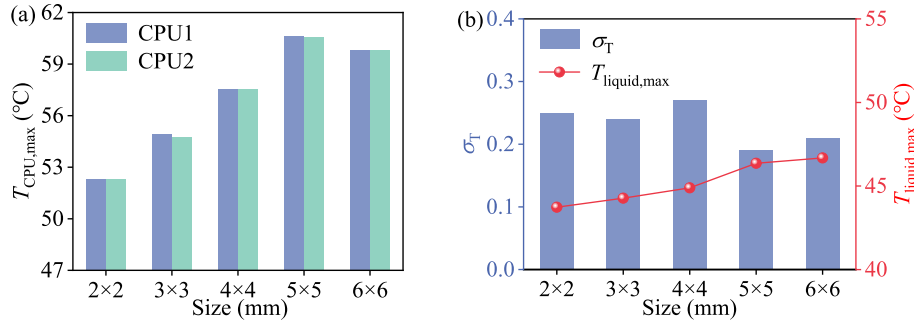


Fig. 17. The impact of pin-fin sizes on temperature characteristics. (a) Peak temperature of CPUs; (b) Temperature uniformity index and peak temperature of coolants.

index and the peak temperature of coolants within the SPILC cabinet are at their lowest when the jet angle is 90°. This is also due to the best cooling when the jet direction is directly facing the CPU heatsink, thereby reducing the thermal cascade effect and promoting a more uniform temperature distribution of the coolant inside jet-assisted SPILC cabinets.

Fig. 16 analyzes the influence of jet angles on the pressure force loss of jet-assisted SPILC cabinets. Obviously, the pressure force loss gradually increases with the enlargement of the jet angle, reaching a maximum at a jet angle of 120°. Data analysis indicates that at a jet angle of 120°, the pressure force loss increases by 2.1 % compared to when it is at 30°. However, when the jet angle increases to 150°, the pressure force loss shows a decrease trend. This is attributed to the fact that the velocity component along the main flow direction gradually decreases as the jet angle increases from 30° to 90°, while the component perpendicular to the main flow gradually increases, leading to a gradual increase in flow resistance. When the jet angle increases from 90° to 120°, the velocity component in the reverse direction of the main flow gradually increases, and the component perpendicular to the main flow direction gradually decreases. In this context, the change in the velocity component in the reverse direction of the main flow has a predominant effect on flow resistance, causing the pressure force loss to continue to increase. When the jet angle increases from 120° to 150°, the velocity component in the reverse direction of the main flow continues to increase, and the component perpendicular to the main flow continues to decrease. Given this, the change in the component perpendicular to the main flow has a predominant impact on flow resistance, leading to a decrease in pressure force loss.

3.3. Performance enhancement by heatsink optimization

Fig. 17 illustrates the impact of CPU heatsink pin-fin sizes on the temperature response characteristics inside the SPILC cabinet. This study maintains the total cross-sectional area and height of the pin-fins constant, selecting six pin-fin sizes: 2 × 2, 3 × 3, 4 × 4, 5 × 5, and 6 × 6 mm, corresponding to pin-fin gaps of 2, 3.25, 4.22, 5.43, and 6 mm, respectively. In this section, the jet-assisted SPILC cabinet is horizontally arranged with a flow rate ratio of 3/7 and a jet angle of 90°. As plotted in Fig. 17(a), the peak CPU temperature gradually rises as the pin-fin size increases. This indicates that within the range of 2 to 6 mm, the smaller the cross-sectional size of the CPU heatsink pin-fins, the better the cooling effect at the CPU. This is because the total surface area of the heatsink increases as the individual pin-fin size decreases, resulting in an augment in the local thermal exchange at the CPU. Fig. 17(b) shows that as the pin-fin size increases, the temperature uniformity index generally declines. This is because the cooling effect near high heat flux devices (i.e., CPU) worsens as the pin-fin cross-sectional size increases, enhancing the thermal cascade effect and causing a decrease in the coolant temperature uniformity inside jet-assisted SPILC cabinets. The maximum coolant temperature also rises gradually, further indicating

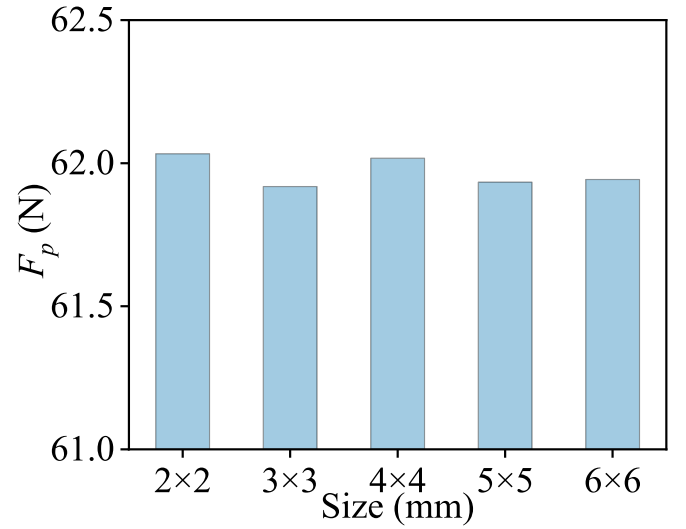


Fig. 18. The impact of pin-fin sizes on the pressure force loss.

that the cooling effect decreases as the pin-fin cross-sectional size increases.

Fig. 18 further reinforces the clarity and precision of the earlier conclusions. The figure unequivocally shows that as the size of the CPU heatsink pin-fins increases, the pressure force loss follows a similar downward trend, reaching the lowest at 3 × 3 mm. This crystal-clear depiction not only confirms the inverse relationship between pin-fin size and flow resistance within the jet-assisted SPILC cabinet but also instills confidence in the findings. The increase in pin-fin size, particularly the expansion of the gaps between them, results in a reduction in the flow resistance of coolants across the heatsink. This, in turn, reduces the overall flow resistance and the pressure force loss of jet-assisted SPILC cabinets.

4. Conclusion

To meet the cooling demands of new-generation high heat-load DCs, this study conducted a meticulous and thorough study on jet-assisted SPILC systems. The flow behaviors and thermal transport characteristics within conventional and jet-assisted SPILC systems were compared. Moreover, efforts were delved into the mechanisms by which cabinet layout, flow rate ratio, and jet angle affect the cooling performance of jet-assisted SPILC systems. In addition, the structure of the CPU heatsink in the jet-assisted SPILC system was optimized. The main conclusions of this study are as follows:

- (1) The overall performance of the jet-assisted SPILC system has significantly improved. Compared with the conventional one, the

CPU temperature in the jet-assisted SPILC system decreases by up to 6.1 %, the local Nusselt number increases by 85.5 %, the coolant temperature uniformity index reduces by 73.1 %, while the pressure force loss is significantly increased.

- (2) For the jet-assisted SPILC system, the CPU cooling performance in the horizontal layout is essentially equivalent to that in the vertical layout. However, the coolant temperature uniformity is significantly enhanced, and the flow resistance is increased.
- (3) Increasing the flow rate ratio significantly improves the cooling performance of jet-assisted SPILC systems and reduces system flow resistance. However, the impact of jet angle on the system's flow and heat transfer characteristics varies. As the jet angle increases, the system's thermal performance improves and is optimal at a jet angle of 90 °. In contrast, the trend in the flow resistance loss is almost the opposite, reaching a maximum at a jet angle of 120 °.
- (4) As the pin-fin size of CPU heatsinks increases, the CPU temperature rises, but the thermal transport uniformity is improved, and the system resistance is reduced. Balancing the trade-off between flow resistance and heat transfer, the overall performance of jet-assisted SPILC systems is optimized when the pin-fin size is 3×3 mm.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] G.I. Meijer, Cooling energy-hungry data centers, *Science* 328 (5976) (2010) 318–319.
- [2] Q. Zhang, Z. Meng, X. Hong, Y. Zhan, J. Liu, J. Dong, T. Bai, J. Niu, M.J. Deen, A survey on data center cooling systems: Technology, power consumption modeling and control strategy optimization, *J. Syst. Architect.* 119 (2021) 102253.
- [3] X.D. Liu, G.Q. Hao, B. Li, Y.P. Chen, Experimental study on the electrohydrodynamic deformation of droplets in a combined DC electric field and shear flow field, *Fund Res.-China* 3 (2023) 274–287.
- [4] Y.P. Chen, Z.L. Deng, Hydrodynamics of droplet passing through a microfluidic T-junction, *J. Fluid Mech* 819 (2017) 401–434.
- [5] W.M. Li, S.Y. Yang, Y.P. Chen, C. Li, Z.K. Wang, Tesla valves and capillary structures-activated thermal regulator, *Nat. Commun.* 14 (2023) 3996.
- [6] X.D. Liu, Y.P. Chen, M.H. Shi, Dynamic performance analysis on start-up of closed-loop pulsating heat pipes (CLPHPs), *Int. J. Therm. Sci.* 65 (2013) 224–233.
- [7] G.H. Zhou, J.Z. Zhou, X.L. Huai, F. Zhou, Y.W. Jiang, A two-phase liquid immersion cooling strategy utilizing vapor chamber heat spreader for data center servers, *Appl. Therm. Eng.* 210 (2022) 118289.
- [8] Y. Koito, Numerical analyses on heat transfer characteristics of ultra-thin heat pipes: Fundamental studies with a three-dimensional thermal-fluid model, *Appl. Therm. Eng.* 148 (2019) 430–437.
- [9] Y.P. Huang, Z.L. Deng, Y.P. Chen, C.B. Zhang, Performance investigation of a biomimetic latent heat thermal energy storage device for waste heat recovery in data centers, *Appl. Energy* 335 (2023) 120745.
- [10] Y.P. Huang, X.D. Liu, Charging and discharging enhancement of a vertical latent heat storage unit by fractal tree-shaped fins, *Renew. Energy* 174 (2021) 199–217.
- [11] Y.P. Huang, D.C. Cao, D.K. Sun, X.D. Liu, Experimental and numerical studies on the heat transfer improvement of a latent heat storage unit using gradient tree-shaped fins, *Int. J. Heat Mass Tran* 182 (2022) 121920.
- [12] M. Azarifar, M. Arik, J. Chang, Liquid cooling of data centers: A necessity facing challenges, *Appl. Therm. Eng.* 247 (2024) 123112.
- [13] J.H. Pu, R.C. Li, Y. Li, H. Zhang, M. Du, N. Hua, X.K. Zhang, The novel stereoscopic cooling plate designs and performance analysis for battery thermal management systems, *Appl. Therm. Eng.* 257 (2024) 124330.
- [14] J. Miao, C. Li, M. Pan, Experimental and numerical analysis of variable cross-section channel liquid cooling plate for server chips thermal management, *Therm. Sci. Eng. Prog.* 49 (2024) 102470.
- [15] S. Qu, K. Duan, Y. Guo, Y. Feng, C. Wang, Z. Xing, Real-time optimization of the liquid-cooled data center based on cold plates under different ambient temperatures and thermal loads, *Appl. Energy* 363 (2024) 123101.
- [16] A. Habibi Khalaj, S.K. Halgamuge, A Review on efficient thermal management of air- and liquid-cooled data centers: From chip to the cooling system, *Appl. Energy* 205 (2017) 1165–1188.
- [17] P.E. Tuma, The merits of open bath immersion cooling of datacom equipment, 2010 26th Annual IEEE Semiconductor Thermal Measurement and Management Symposium (SEMI-THERM), IEEE, 2010, pp. 123–131.
- [18] X. Li, Z. Xu, S. Liu, X. Zhang, H. Sun, Server performance optimization for single-phase immersion cooling data center, *Appl. Therm. Eng.* 224 (2023) 120080.
- [19] M. Muneeswaran, Y. Lin, C. Wang, Performance analysis of single-phase immersion cooling system of data center using FC-40 dielectric fluid, *Int. Commun. Heat Mass* 145 (2023) 106843.
- [20] P. Taddeo, J. Romani, J. Summers, J. Gustafsson, I. Martorell, J. Salom, Experimental and numerical analysis of the thermal behaviour of a single-phase immersion-cooled data centre, *Appl. Therm. Eng.* 234 (2023) 121260.
- [21] M. Hnayno, A. Chehade, H. Klabi, G. Polidori, C. Maalouf, Experimental investigation of a data-centre cooling system using a new single-phase immersion/liquid technique, *Case Stud. Therm. Eng.* 45 (2023) 102925.
- [22] X. Sun, Z. Han, X. Li, Simulation study on cooling effect of two-phase liquid-immersion cabinet in data center, *Appl. Therm. Eng.* 207 (2022) 118142.
- [23] X. Chen, T. Ding, H. Cao, H. Ding, Z. Li, Flow boiling heat transfer mechanisms and flow characteristics of pump-driven two-phase flow systems used in data center cooling, *Appl. Therm. Eng.* 220 (2023) 119642.
- [24] B.B. Kanbur, C. Wu, S. Fan, W. Tong, F. Duan, Two-phase liquid-immersion data center cooling system: Experimental performance and thermoeconomic analysis, *Int. J. Refrig.* 118 (2020) 290–301.
- [25] H. Wang, X. Yuan, K. Zhang, X. Lang, H. Chen, H. Yu, S. Li, Performance evaluation and optimization of data center servers using single-phase immersion cooling, *Int. J. Heat Mass Tran.* 221 (2024) 125057.
- [26] Y. Wang, C. Li, X. Wen, W. Cai, Y. Jiang, C. Wen, Y. Wang, L. Hu, H. Yu, H. Zhu, H. Guo, D. Liu, Experimental studies on two-phase immersion liquid cooling for Li-ion battery thermal management, *J. Energy Storage* 72 (2023) 108748.
- [27] Y.P. Huang, J.L. Ge, Y.P. Chen, C.B. Zhang, Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers, *Int. J. Heat Mass Tran.* 207 (2023) 124023.
- [28] Y.P. Huang, B. Liu, S.J. Xu, C.J. Bao, Y.F. Zhong, C.B. Zhang, Experimental study on the immersion liquid cooling performance of high-power data center servers, *Energy* 297 (2024) 131195.
- [29] X. Chen, Y.P. Huang, S.J. Xu, C.J. Bao, Y.F. Zhong, Y.P. Chen, C.B. Zhang, Thermal performance evaluation of electronic fluorinated liquids for single-phase immersion liquid cooling, *Int. J. Heat Mass Tran.* 220 (2024) 124951.
- [30] S. Liu, Z. Xu, Z. Wang, X. Li, H. Sun, X. Zhang, H. Zhang, Optimization and comprehensive evaluation of liquid cooling tank for single-phase immersion cooling data center, *Appl. Therm. Eng.* 245 (2024) 122864.
- [31] D. Ni, F. Yu, CFD simulation study of flow equalisation plate model in single-phase immersion liquid cooling for servers, *Therm. Sci. Eng. Prog.* 47 (2024) 102268.
- [32] H. Shrigondekar, Y. Lin, C. Wang, Investigations on performance of single-phase immersion cooling system, *INT J HEAT MASS TRAN* 206 (2023) 123961.
- [33] W. Qi, Y. Xu, H. Li, Performance analysis and design optimization of avionics devices with single-phase submerged jet cooling, *Appl. Therm. Eng.* 231 (2023) 120993.
- [34] Z. Xue, Y. Yan, Z. He, J. You, C. Zhang, Performance analysis of trade-off for thermal and hydraulic characteristics for liquid-jet-cooled heat sink, *Int. J. Therm. Sci.* 203 (2024) 109139.
- [35] H. Jouhara, A.J. Robinson, Experimental investigation of small diameter two-phase closed thermosyphons charged with water, FC-84, FC-77 and FC-3283, *Appl. Therm. Eng.* 30 (2010) 201–211.
- [36] P. Naphon, S. Klangchart, S. Wongwises, Numerical investigation on the heat transfer and flow in the mini-fin heat sink for CPU, *INT COMMUN HEAT MASS* 36 (2009) 834–840.
- [37] H.J. Xiang, Y.P. Wang, L. Zhu, X.Y. Han, Y. Sun, Z.J. Zhao, 3D numerical simulation on heat transfer performance of a cylindrical liquid immersion solar receiver, *Energ. Convers. Manage.* 64 (2012) 97–105.